

Self-contact frictional jamming of helical ribbons

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Abstract

A paper yoyo is a ribbon wound around itself in self-contact and pulled apart along its winding axis. We derive a quasi-static continuum model of the resulting force-displacement relation $F(x)$ by deriving and assembling five independently derived components: the winding kinematics, the three-dimensional stress balance, constitutive relations and compatibility, Coulomb friction at the self-contact interfaces, and elastic resistance to bending away from the ribbon’s natural curl. The assembly produces two coupled terms of $F(x)$. First is the conjugate force of the stored bending energy, which is fully determined by the ribbon’s geometry and elastic moduli. The second is the friction-driven membrane stress, which requires a normalization that the friction equations cannot supply on their own; we show that the ribbon’s own arc-length constraint supplies it, and solve the resulting coupled system numerically. Along the way we identify a “locking” phenomena at which it becomes more energetically favorable to stretch the material rather than kinematically tighten the coil.

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1 The system, and the assumptions

The system is a long constant-width rectangular ribbon of small but finite thickness, wrapped around itself at a uniform pitch. One end of the ribbon is fixed to prevent rigid-body motion; the other is displaced by a distance x along the winding axis, which is taken to be the z -axis (Figure 1).

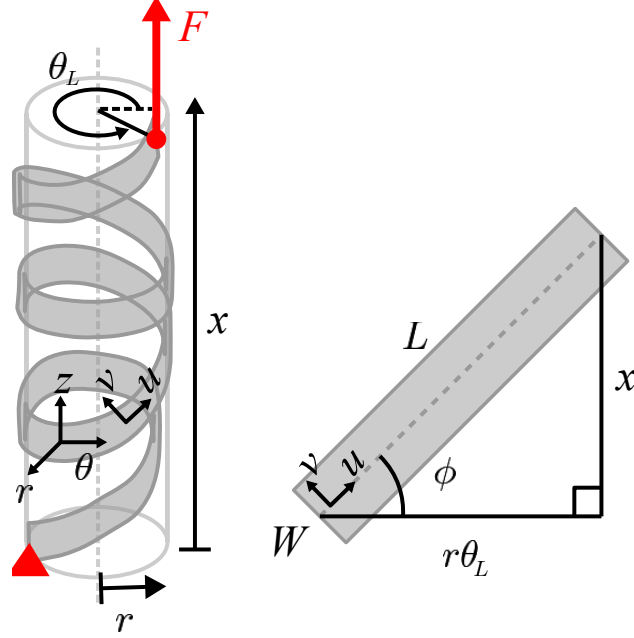


Figure 1: The wound ribbon. One end is held fixed; the other (“active”) end is displaced by x along the winding axis.

1.1 Geometric and material constants

The geometric and material constants are

- L : the length of the ribbon in its natural, unstressed state.
- W : its width, with $W \ll L$.
- t : its thickness, with $t \ll W$.
- θ_L : the total wind angle. This is *fixed* by the boundary conditions: the two ends cannot rotate about z relative to one another, only translate along it, so the number of turns is a topological property of how the ribbon was initially wound and does not change as it is pulled. $\theta_L = 2\pi n$ where n is the (not necessarily integer) number of turns.
- E, ν : Young’s modulus and Poisson ratio of the ribbon’s material.
- μ : the Coulomb friction coefficient for sheet-on-sheet contact.

1.2 Assumptions

It is worth putting these in one place, since several of the model’s conclusions turn on them.

1. **Quasi-static loading.** Inertia is neglected; the state at each x is an equilibrium state.
2. **Uniform pitch angle ϕ .** The angle the local ribbon direction makes with the (r, θ) plane is the same everywhere on the sheet at a given x . This is a geometric simplification that keeps the surface developable (Section 2.3).

3. **Uniform coil radius r .** At a given x , every material point sits at the same radius from the winding axis. This is another geometric simplification: successive turns really differ in radius by one thickness, so r varies by $t\theta_L/2\pi$ across the ribbon. The fractional variation is $t\theta_L/(2\pi r)$, small for a coil wound with a central hole $r \gg t$. We retain the effect where it matters—it sets the packing floor of Section 2.4—but not in the stress field.
4. **Linear elasticity, small strain, and plane stress.** The sheet is Hookean, and thin enough that the through-thickness stress state is plane.
5. **Linear variation of every stress component through the thickness.** Because t is small, the variation of values across the thickness can be approximated as linear. Therefore, knowing a component's two surface values is then equivalent to knowing its full profile by linear interpolation.
6. **Coulomb friction with a velocity-aligned direction.** At each contact point between successive layers, the shear traction has magnitude μ times the normal pressure and opposes the local relative sliding velocity. This is a sliding (kinetic) law, so it presumes the interface is slipping rather than stuck. Section 5.2 shows the relative slip velocity is nonzero for every $x > 0$, so the presumption is self-consistent, with the single exception of $x = 0$ exactly.
7. **Contact transmits pressure and friction only, with no adhesion.** Two surfaces can push on each other but not pull, so $\sigma_{rr} \leq 0$. We state this as a modelling requirement but do *not* enforce it as an inequality constraint; Section 9.3 reports where the solution violates it.

2 Kinematics

This section derives the system geometry before introducing forces.

2.1 Variables

The single defining state variable is x . All other variables describe the system at one frozen value of x , or about how it changes as x varies.

The key variables to describe the geometry of a given state are the coil radius r and the pitch angle ϕ . As mentioned in Section 1.2, we assume both are uniform across the ribbon at a given x . The only other geometric variable is the arc length of a material line along the ribbon (i.e., the length of the ribbon), which we denote Λ . It is related to r and x by the Pythagorean relation

$$\Lambda = \sqrt{(r\theta_L)^2 + x^2}. \quad (1)$$

If the ribbon carries a mean axial strain $\epsilon := \epsilon_{uu}$, that same material line has length $L(1 + \epsilon)$, so

$$\Lambda = L(1 + \epsilon), \quad r\theta_L = \sqrt{L^2(1 + \epsilon)^2 - x^2}. \quad (2)$$

By unrolling this strained ribbon into a the hypotenuse of a right triangle, we can relate

$$\sin \phi = \frac{x}{\Lambda} \quad \text{and} \quad \cos \phi = \frac{r \theta_L}{\Lambda} \quad (3)$$

Then,

$$x = L(1 + \epsilon) \sin \phi \quad \xrightarrow{\epsilon \rightarrow 0} \quad x = L \sin \phi, \quad r \theta_L = L \cos \phi. \quad (4)$$

Observe that for a fixed ϵ , increasing x forces ϕ to increase, which causes r to decrease. In other words, the coil tightens as it is pulled. It is worth being explicit that this tightening is a purely kinematic rearrangement which costs no stretching: Section 2.3 shows the deformation is an exact isometry.

We now define the following nondimensional variables:

$$\boxed{r_{\text{nat}} := \frac{L}{\theta_L}, \quad \rho := \frac{r}{r_{\text{nat}}}, \quad \chi := \frac{x}{L}, \quad \lambda := \frac{\Lambda}{L} = \sqrt{\rho^2 + \chi^2} = 1 + \epsilon.} \quad (5)$$

- r_{nat} is the coil radius at the resting state at $x = 0$, or in other words, the radius of the material's natural curvature. Changing the curvature of the material produces bending stress. It has units of length.
- ρ is the current coil radius normalized by r_{nat} .
- χ is the current axial displacement normalized by the ribbon length.
- λ is the current arc length normalized by the ribbon length.

2.2 Position vector and local frame

The position of a point on the ribbon can be described either by surface coordinates (u, v) on the unrolled rectangle— $u \in [0, L]$ along its length, $v \in [-W/2, W/2]$ across its width—or by cylindrical coordinates (r, θ, z) in space. The through-thickness coordinate is $\zeta \in [-t/2, t/2]$, measured from the mid-surface along the outward radial direction $\hat{r}$, so a general material point sits at $\vec{p}(u, v, x) + \zeta \hat{r}$.

Uniform pitch angle ϕ means the local ribbon direction is inclined at ϕ to the (r, θ) plane, so

$$\frac{\partial z}{\partial u} = \sin \phi, \quad \frac{\partial z}{\partial v} = \cos \phi, \quad (6)$$

and, at uniform radius, the circumferential relations are

$$r \frac{\partial \theta}{\partial u} = \cos \phi, \quad r \frac{\partial \theta}{\partial v} = -\sin \phi. \quad (7)$$

With r uniform, Eqs. (6)–(7) integrate directly to obtain

$$\boxed{\vec{p}(u, v, x) = \begin{bmatrix} r \\ \theta \\ z \end{bmatrix} = \begin{bmatrix} r(x) \\ \frac{u \cos \phi - v \sin \phi}{r(x)} \\ u \sin \phi + v \cos \phi \end{bmatrix}}. \quad (8)$$

Differentiating in the cylindrical basis, using $\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta}$, $\frac{\partial \hat{\theta}}{\partial \theta} = -\hat{r}$ and $\frac{\partial \hat{z}}{\partial \theta} = 0$,

$$\frac{\partial \vec{p}}{\partial u} = \cos \phi \hat{\theta} + \sin \phi \hat{z}, \quad \frac{\partial \vec{p}}{\partial v} = -\sin \phi \hat{\theta} + \cos \phi \hat{z}, \quad (9)$$

which are exactly orthonormal, so we may set $\hat{u} := \frac{\partial \vec{p}}{\partial u}$ and $\hat{v} := \frac{\partial \vec{p}}{\partial v}$ with no approximation.

2.3 Curvature and developability

Differentiating $\hat{u}$ and $\hat{v}$ once more and reading off the components along $\hat{r}$ gives the second fundamental form:

$$\boxed{\kappa_{uu} = \frac{\cos^2 \phi}{r}, \quad \kappa_{uv} = \frac{\sin \phi \cos \phi}{r}, \quad \kappa_{vv} = \frac{\sin^2 \phi}{r}}, \quad (10)$$

Together with the frame-rotation rates needed in Section 3, the Gaussian curvature vanishes:

$$\frac{\partial \hat{r}}{\partial u} = \kappa_{uu} \hat{u} - \kappa_{uv} \hat{v}, \quad \frac{\partial \hat{r}}{\partial v} = -\kappa_{uv} \hat{u} + \kappa_{vv} \hat{v}. \quad (11)$$

$$\boxed{K = \kappa_{uu}\kappa_{vv} - \kappa_{uv}^2 = \frac{\sin^2 \phi \cos^2 \phi}{r^2} - \frac{\sin^2 \phi \cos^2 \phi}{r^2} = 0}. \quad (12)$$

Therefore, the surface is always developable.

2.4 The minimum radius

The one place the finite thickness of the stack must be retained is the limit on how far the coil can tighten. Successive turns are separated radially by one thickness, so over θ_L of winding the coil occupies a radial band of extent $t\theta_L/2\pi$. The mean radius cannot fall below half that band without the innermost turn closing on the axis (that is, radius exactly zero), giving a hard floor

$$\boxed{\rho_{\text{jam}} = \frac{t\theta_L^2}{4\pi L} = \frac{tn}{2r_{\text{nat}}}, \quad n = \frac{\theta_L}{2\pi}}. \quad (13)$$

The second form reads transparently: ρ_{jam} is half the coil's total radial build measured against its initial radius. A coil wound with a large central hole would have $tn \ll r_{\text{nat}}$ and $\rho_{\text{jam}} \ll 1$; one with a tight initial winding would have $tn \rightarrow r_{\text{nat}}$ and $\rho_{\text{jam}} \rightarrow 1/2$.

3 Stress balance

3.1 Notation

At a point (u, v, ζ) the material carries the full Cauchy stress

$$\sigma = \begin{pmatrix} \sigma_{uu} & \sigma_{uv} & \sigma_{ur} \\ \sigma_{uv} & \sigma_{vv} & \sigma_{vr} \\ \sigma_{ur} & \sigma_{vr} & \sigma_{rr} \end{pmatrix}, \quad (14)$$

with six independent components: the in-plane stresses $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$; the transverse shears σ_{ur}, σ_{vr} , by which the ribbon transmits load sideways through its thickness; and σ_{rr} , the contact pressure between layers.

Two kinds of through-thickness reduction recur. Thickness averages are written with a bar. Under the linear-profile assumption (assumption 5, Section 1.2) the average of a component equals the mean of its two surface values:

$$\bar{\sigma}_{ij} := \frac{1}{t} \int_{-t/2}^{t/2} \sigma_{ij} d\zeta = \frac{1}{2} \left[\sigma_{ij} \left(\zeta = \frac{t}{2} \right) + \sigma_{ij} \left(\zeta = -\frac{t}{2} \right) \right]. \quad (15)$$

We also define the difference of the two surface values using a Δ :

$$\Delta\sigma_{ij} := \sigma_{ij} \left(\zeta = \frac{t}{2} \right) - \sigma_{ij} \left(\zeta = -\frac{t}{2} \right). \quad (16)$$

Finally, the two surface values of the contact pressure σ_{rr} get their own names:

$$\sigma_{rr}^{\pm} := \sigma_{rr}(\zeta = \pm \frac{t}{2}) \quad (17)$$

3.2 Force balance on a curved element

Perform a force balance on a volume element $du \times dv \times d\zeta$. On its u -face the traction is $\vec{t}_u = \sigma_{uu}\hat{u} + \sigma_{uv}\hat{v} + \sigma_{ur}\hat{r}$, and similarly $\vec{t}_v$ and $\vec{t}_\zeta$ on the other two. The faces of the element are approximately rectangular because $t \ll r$, so

$$\left(\frac{\partial \vec{t}_u}{\partial u} + \frac{\partial \vec{t}_v}{\partial v} + \frac{\partial \vec{t}_\zeta}{\partial \zeta} \right) du dv d\zeta = 0. \quad (18)$$

The basis vectors $\hat{u}$, $\hat{v}$, and $\hat{r}$ themselves depend on position (Eq. 11), so differentiating the tractions produces curvature-weighted terms beyond the plain derivatives of the stress components. For example,

$$\frac{\partial}{\partial u} (\sigma_{uu}\hat{u}) = \frac{\partial \sigma_{uu}}{\partial u} \hat{u} + \sigma_{uu} \frac{\partial \hat{u}}{\partial u} = \frac{\partial \sigma_{uu}}{\partial u} \hat{u} - \kappa_{uu} \sigma_{uu} \hat{r}, \quad (19)$$

The extra term expresses the fact that the element is not planar, so in-plane stresses are slightly rotated into the out-of-plane direction, and vice versa. Expanding every term this way and sorting

by component gives three scalar equations:

$$\hat{u} : \quad \frac{\partial \sigma_{uu}}{\partial u} + \frac{\partial \sigma_{uv}}{\partial v} + \frac{\partial \sigma_{ur}}{\partial \zeta} + \kappa_{uu} \sigma_{ur} - \kappa_{uv} \sigma_{vr} = 0 \quad (20)$$

$$\hat{v} : \quad \frac{\partial \sigma_{uv}}{\partial u} + \frac{\partial \sigma_{vv}}{\partial v} + \frac{\partial \sigma_{vr}}{\partial \zeta} - \kappa_{uv} \sigma_{ur} + \kappa_{vv} \sigma_{vr} = 0 \quad (21)$$

$$\hat{r} : \quad \frac{\partial \sigma_{ur}}{\partial u} + \frac{\partial \sigma_{vr}}{\partial v} + \frac{\partial \sigma_{rr}}{\partial \zeta} - \kappa_{uu} \sigma_{uu} + 2\kappa_{uv} \sigma_{uv} - \kappa_{vv} \sigma_{vv} = 0 \quad (22)$$

Equilibrium alone provides three equations for these six unknown stress components. The remaining relations come from friction (Section 5), which supplies the surface values of σ_{ur} , σ_{vr} , σ_{rr} , and from compatibility (Section 4).

3.3 Zeroth thickness moment

Averaging Eqs. (20)–(22) over the thickness, and using $\frac{1}{t} \int \frac{\partial \sigma_{ur}}{\partial \zeta} d\zeta = \frac{1}{t} \Delta \sigma_{ur}$,

$$\frac{\partial \bar{\sigma}_{uu}}{\partial u} + \frac{\partial \bar{\sigma}_{uv}}{\partial v} + \frac{1}{t} \Delta \sigma_{ur} + \kappa_{uu} \bar{\sigma}_{ur} - \kappa_{uv} \bar{\sigma}_{vr} = 0 \quad (23)$$

$$\frac{\partial \bar{\sigma}_{uv}}{\partial u} + \frac{\partial \bar{\sigma}_{vv}}{\partial v} + \frac{1}{t} \Delta \sigma_{vr} - \kappa_{uv} \bar{\sigma}_{ur} + \kappa_{vv} \bar{\sigma}_{vr} = 0 \quad (24)$$

$$\frac{\partial \bar{\sigma}_{ur}}{\partial u} + \frac{\partial \bar{\sigma}_{vr}}{\partial v} + \frac{1}{t} \Delta \sigma_{rr} - \kappa_{uu} \bar{\sigma}_{uu} + 2\kappa_{uv} \bar{\sigma}_{uv} - \kappa_{vv} \bar{\sigma}_{vv} = 0 \quad (25)$$

These are still exact consequences of Eqs. (20)–(22); no term has been dropped.

3.4 First order moment balance

Multiplying by ζ and integrating Eqs. (20)–(22) gives an independent statement—the moment balance—which Section ?? needs. Define the bending moments and the first moment of the transverse shear,

$$M_{ij} := \int_{-t/2}^{t/2} \zeta \sigma_{ij} d\zeta \quad (ij \in \{uu, uv, vv\}), \quad \Xi_{ir} := \int_{-t/2}^{t/2} \zeta \sigma_{ir} d\zeta \quad (i \in \{u, v\}). \quad (26)$$

Multiplying Eq. (20) by ζ , integrating, and integrating the $\frac{\partial}{\partial \zeta}$ term by parts,

$$\int_{-t/2}^{t/2} \zeta \frac{\partial \sigma_{ur}}{\partial \zeta} d\zeta = \frac{t}{2} [\sigma_{ur}(\frac{t}{2}) + \sigma_{ur}(-\frac{t}{2})] - t \bar{\sigma}_{ur} = 0, \quad (27)$$

where the final equality is exactly the linear-profile identity. So the boundary term cancels identically and

$$\frac{\partial M_{uu}}{\partial u} + \frac{\partial M_{uv}}{\partial v} + \kappa_{uu} \Xi_{ur} - \kappa_{uv} \Xi_{vr} = 0 \quad (28)$$

with the companion from Eq. (21),

$$\frac{\partial M_{uv}}{\partial u} + \frac{\partial M_{vv}}{\partial v} - \kappa_{uv}\Xi_{ur} + \kappa_{vv}\Xi_{vr} = 0 \quad (29)$$

This is the shell analogue of the elementary beam relation between a moment gradient and a transverse shear force. Under the uniform- r , uniform- ϕ assumptions of this model the moments turn out to depend on x alone and not on (u, v) (Section 6.2), so both gradient terms vanish. Eqs. (28)–(29) reduce to the homogeneous pair

$$\boxed{\kappa_{uu}\Xi_{ur} - \kappa_{uv}\Xi_{vr} = 0} \quad \boxed{-\kappa_{uv}\Xi_{ur} + \kappa_{vv}\Xi_{vr} = 0} \quad (30)$$

whose determinant is $\kappa_{uu}\kappa_{vv} - \kappa_{uv}^2 = K = 0$ by Eq. (12). The pair is therefore singular: it neither forces the transverse shear moments to vanish nor fixes them. A spatially uniform curvature mismatch, in a developable geometry, simply carries no information about the transverse shear. This is why bending turns out not to drive the contact pressure (Section ??).

4 Constitutive relations and compatibility

4.1 Hooke's law

Because the ribbon is thin and its long edges are traction-free, the in-plane state is plane *stress*. For a linear isotropic material,

$$\bar{\epsilon}_{uu} = \frac{1}{E}(\bar{\sigma}_{uu} - \nu \bar{\sigma}_{vv}), \quad \bar{\epsilon}_{vv} = \frac{1}{E}(\bar{\sigma}_{vv} - \nu \bar{\sigma}_{uu}), \quad \bar{\epsilon}_{uv} = \frac{1+\nu}{E} \bar{\sigma}_{uv}, \quad (31)$$

inverted where convenient as

$$\bar{\sigma}_{uu} = \frac{E}{1-\nu^2}(\bar{\epsilon}_{uu} + \nu \bar{\epsilon}_{vv}), \quad \bar{\sigma}_{vv} = \frac{E}{1-\nu^2}(\bar{\epsilon}_{vv} + \nu \bar{\epsilon}_{uu}), \quad \bar{\sigma}_{uv} = \frac{E}{1+\nu} \bar{\epsilon}_{uv}. \quad (32)$$

4.2 Compatibility, with no curvature source

Equilibrium constrains forces to balance; it says nothing about whether the resulting strain field corresponds to an actual continuous displacement of an unbroken sheet. For a strain field on a surface of Gaussian curvature K , the linearized (von Kármán) compatibility relation carries $-K$ on its right-hand side. Here $K = 0$ exactly (Eq. 12), so the relation reduces to the flat Saint-Venant statement with no source term:

$$\boxed{\frac{\partial^2 \bar{\epsilon}_{uu}}{\partial v^2} + \frac{\partial^2 \bar{\epsilon}_{vv}}{\partial u^2} - 2 \frac{\partial^2 \bar{\epsilon}_{uv}}{\partial u \partial v} = 0.} \quad (33)$$

It is worth stating plainly what this rules out. One might hope that the tightening of the coil geometrically *forces* some membrane strain, which would supply the stress field with a source and so determine its magnitude. It does not: the tightening is isometric, so Eq. (33) is homogeneous, and the question of what sets the magnitude of the friction-driven stress is left open until Section 8.3.

Substituting Eq. (31) into Eq. (33) and multiplying through by E gives:

$$\boxed{\frac{\partial^2 \bar{\sigma}_{uu}}{\partial v^2} - \nu \frac{\partial^2 \bar{\sigma}_{vv}}{\partial v^2} + \frac{\partial^2 \bar{\sigma}_{vv}}{\partial u^2} - \nu \frac{\partial^2 \bar{\sigma}_{uu}}{\partial u^2} - 2(1 + \nu) \frac{\partial^2 \bar{\sigma}_{uv}}{\partial u \partial v} = 0.} \quad (34)$$

4.3 Equivalent displacement form

Eq. (34) is the awkward member of the governing set, and there is an exactly equivalent formulation that avoids it. Let $a(u, v)$ and $b(u, v)$ be the in-plane displacements along $\hat{u}$ and $\hat{v}$, so that

$$\bar{\epsilon}_{uu} = \frac{\partial a}{\partial u}, \quad \bar{\epsilon}_{vv} = \frac{\partial b}{\partial v}, \quad \bar{\epsilon}_{uv} = \frac{1}{2} \left(\frac{\partial a}{\partial v} + \frac{\partial b}{\partial u} \right). \quad (35)$$

Compatibility is then satisfied *identically*: it is nothing more than the statement that a displacement field exists, which Eq. (35) presumes. Writing the thickness-averaged in-plane equilibrium equations (23)–(24) with everything that is not an in-plane stress gradient moved to the right,

$$\frac{\partial \bar{\sigma}_{uu}}{\partial u} + \frac{\partial \bar{\sigma}_{uv}}{\partial v} = f_u^b, \quad \frac{\partial \bar{\sigma}_{uv}}{\partial u} + \frac{\partial \bar{\sigma}_{vv}}{\partial v} = f_v^b, \quad (36)$$

the pair (f_u^b, f_v^b) acts as an in-plane body force. Substituting Eqs. (32)–(35) turns Eq. (36) into the Navier equations:

$$\frac{E}{1 - \nu^2} \frac{\partial^2 a}{\partial u^2} + \frac{E}{2(1 + \nu)} \frac{\partial^2 a}{\partial v^2} + \left[\frac{E\nu}{1 - \nu^2} + \frac{E}{2(1 + \nu)} \right] \frac{\partial^2 b}{\partial u \partial v} = f_u^b, \quad (37)$$

$$\frac{E}{2(1 + \nu)} \frac{\partial^2 b}{\partial u^2} + \frac{E}{1 - \nu^2} \frac{\partial^2 b}{\partial v^2} + \left[\frac{E\nu}{1 - \nu^2} + \frac{E}{2(1 + \nu)} \right] \frac{\partial^2 a}{\partial u \partial v} = f_v^b. \quad (38)$$

These are two equations for two unknowns, second order, with the body force appearing undifferentiated. This is the form the numerical work of Section 9 uses, and Section 8.2 verifies that it is equivalent to Eq. (34).

5 Friction

5.1 Overlapping neighbours

Because the ribbon is wound around itself, an element at (u, v) can be in contact with another element one full turn away along the ribbon: at $(u^\pm, v^\pm)$, with equal z and θ differing by $\pm 2\pi$. Solving Eq. (8) for those two conditions,

$$\boxed{u^\pm = u \pm 2\pi r \cos \phi, \quad v^\pm = v \mp 2\pi r \sin \phi.} \quad (39)$$

Neglecting edge effects where $u^\pm$ falls outside $[0, L]$, an inner neighbour exists exactly when v^- lies within the ribbon's width, and likewise for the outer:

$$\text{inner neighbour present} \iff v \leq \frac{W}{2} - 2\pi r \sin \phi, \quad (40)$$

$$\text{outer neighbour present} \iff v \geq -\frac{W}{2} + 2\pi r \sin \phi. \quad (41)$$

5.2 Slip kinematics and the friction direction

Coulomb friction opposes relative sliding, so we need the relative velocity of two elements in contact as x advances. Differentiating Eq. (8) at fixed (u, v) ,

$$\frac{\partial r}{\partial x} = \frac{dr}{dx}, \quad \frac{\partial \theta}{\partial x} = -\frac{\theta}{r} \frac{dr}{dx} - \frac{u \sin \phi + v \cos \phi}{r} \frac{d\phi}{dx}, \quad \frac{\partial z}{\partial x} = (u \cos \phi - v \sin \phi) \frac{d\phi}{dx}. \quad (42)$$

Subtracting the same quantities at the inner contact partner (Eq. 39), converting the angular part to an arc-length rate by multiplying by r , and using $\theta(u^-, v^-) - \theta(u, v) = -2\pi$ exactly, the position-dependent pieces cancel and

$$\Delta v_\theta = -2\pi \frac{dr}{dx}, \quad \Delta v_z = 2\pi r \frac{d\phi}{dx}. \quad (43)$$

Both are independent of (u, v) : every contact in the coil slides at the same rate. Projecting onto the surface frame using the rotation by ϕ ,

$$\vec{v}_{\text{slip}} = 2\pi \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} -\frac{dr}{dx} \\ r \frac{d\phi}{dx} \end{pmatrix}, \quad (44)$$

and defining (f_u, f_v) as the unit vector along the slip direction,

$$f_u = \frac{-\frac{dr}{dx} \cos \phi + r \frac{d\phi}{dx} \sin \phi}{\sqrt{\left(\frac{dr}{dx}\right)^2 + r^2 \left(\frac{d\phi}{dx}\right)^2}}, \quad f_v = \frac{\frac{dr}{dx} \sin \phi + r \frac{d\phi}{dx} \cos \phi}{\sqrt{\left(\frac{dr}{dx}\right)^2 + r^2 \left(\frac{d\phi}{dx}\right)^2}}. \quad (45)$$

In the inextensible limit this collapses completely. There $r = r_{\text{nat}} \cos \phi$, so $\frac{dr}{dx} = -r_{\text{nat}} \sin \phi \frac{d\phi}{dx}$, and on substituting into Eq. (45) both $\frac{d\phi}{dx}$ and r_{nat} cancel between numerator and denominator, leaving

$$\boxed{(f_u, f_v) = (\sin 2\phi, \cos 2\phi)}. \quad (46)$$

We use this form from here on, and note the one place it degenerates: at $x = 0$ exactly, $\frac{dr}{dx} = 0$ and $\phi = 0$, so Eq. (46) gives $(f_u, f_v) = (0, 1)$ — the only sliding at the very start of the pull is axial, with no circumferential component at all. Since $\Delta v_z = 2\pi r \frac{d\phi}{dx} \neq 0$ there, sliding is nonetheless active for every $x > 0$, which is what justifies using a kinetic friction law throughout (assumption 6, Section 1.2).

5.3 Friction magnitude

The friction exerted on an element by the layer inside it resists that element's forward motion, and its components are the shear stresses on the inner ($-\hat{r}$) face. Coulomb's law makes the magnitude μ times the normal pressure $-\sigma_{rr}^-$, so

$$\sigma_{ur}\left(-\frac{t}{2}\right) = -\mu \sigma_{rr}^- f_u, \quad \sigma_{vr}\left(-\frac{t}{2}\right) = -\mu \sigma_{rr}^- f_v. \quad (47)$$

The friction from the outer layer is the exact opposite in physical direction; because the outward normal of that face is $+\hat{r}$ rather than $-\hat{r}$, the sign convention for shear stresses makes the stress components take the same form,

$$\sigma_{ur}\left(\frac{t}{2}\right) = -\mu \sigma_{rr}^+ f_u, \quad \sigma_{vr}\left(\frac{t}{2}\right) = -\mu \sigma_{rr}^+ f_v. \quad (48)$$

Each pair holds only where the corresponding neighbour is present (Eqs. 40–41); where it is absent, both the shear and the pressure that would drive it vanish. Taking the mean and the difference of Eqs. (47)–(48),

$$\boxed{\bar{\sigma}_{ur} = -\mu \bar{\sigma}_{rr} f_u, \quad \bar{\sigma}_{vr} = -\mu \bar{\sigma}_{rr} f_v,} \quad (49)$$

$$\Delta\sigma_{ur} = -\mu \Delta\sigma_{rr} f_u, \quad \Delta\sigma_{vr} = -\mu \Delta\sigma_{rr} f_v. \quad (50)$$

5.4 Layer-contact recursion

By Newton's third law, two surfaces in contact carry equal and opposite tractions; since their normals are opposite, the stress components are equal. It suffices to equate the normal pressures since the shears are tied to them by Eq. (49). This gives the recursive relationship

$$\text{outer surface: } \sigma_{rr}^+(u, v) = \begin{cases} \sigma_{rr}^-(u^+, v^+) & \text{outer neighbour present} \\ 0 & \text{otherwise} \end{cases} \quad (51)$$

$$\text{inner surface: } \sigma_{rr}^-(u, v) = \begin{cases} \sigma_{rr}^+(u^-, v^-) & \text{inner neighbour present} \\ 0 & \text{otherwise} \end{cases} \quad (52)$$

Propagated far enough in u , Eq. (51) reaches the outermost wound layer near $u = L$, past which no material is wound at all, so $\sigma_{rr}^+ = 0$ there. That is the base case that starts the recursion, and it is a geometric fact rather than an imposed boundary condition: nothing is wound outside the last turn, independent of how the tip happens to be gripped.

Eqs. (51)–(52) are exact, and the numerical work of Section 9 uses them in that form. However, for analytical work, it is convenient to Taylor-expand in the shift of Eq. (39), which in the case where both neighbours are present gives the differential relation

$$\boxed{\Delta\sigma_{rr} = 2\pi r \cos \phi \frac{\partial \bar{\sigma}_{rr}}{\partial u} - 2\pi r \sin \phi \frac{\partial \bar{\sigma}_{rr}}{\partial v}.} \quad (53)$$

This is a first-order transport equation: it advects $\bar{\sigma}_{rr}$ along the direction $(\cos \phi, -\sin \phi)$ in the

(u, v) plane, driven by $\Delta\sigma_{rr}$. Its characteristic direction is the ribbon's own winding direction, as it must be, since the recursion steps from one turn to the next.

6 Bending resistance

We now account for the material's resistance to bending. This component is an independent physical effect, and can be fully described without introducing any undetermined constants.

6.1 Natural curvature and the mismatch

A sheet that has been wound and left alone acquires a preferred curl. We take that natural state to be a cylinder of radius $r_{\text{nat}} = L/\theta_L$ at zero pitch, i.e., the curvature triple $\kappa_{\text{nat}} = [1/r_{\text{nat}}, 0, 0]$ in the ordering $[\kappa_{uu}, \kappa_{uv}, \kappa_{vv}]$. The mismatch between actual and natural curvature is then

$$\boxed{\Delta\kappa_{uu} = \frac{\cos^2 \phi}{r} - \frac{1}{r_{\text{nat}}}, \quad \Delta\kappa_{uv} = \frac{\sin \phi \cos \phi}{r}, \quad \Delta\kappa_{vv} = \frac{\sin^2 \phi}{r}}, \quad (54)$$

and it satisfies two identities:

$$\Delta\kappa_{uu} + \Delta\kappa_{vv} = \frac{1}{r} - \frac{1}{r_{\text{nat}}}, \quad \Delta\kappa_{uu}\Delta\kappa_{vv} - \Delta\kappa_{uv}^2 = -\frac{\sin^2 \phi}{r r_{\text{nat}}}. \quad (55)$$

The first is immediate from $\cos^2 \phi + \sin^2 \phi = 1$. In the second, the cross terms cancel, leaving $-\sin^2 \phi / (r r_{\text{nat}})$, which is nonzero for any $\phi \neq 0$. Note that this is the determinant of the *mismatch*, not of the curvature itself, so there is no contradiction with $K = 0$ (Eq. 12).

At $x = 0$, we have $\phi = 0$ and $\rho = 1$, so all three components of Eq. (54) vanish and the sheet is unstressed as intended. Note also that each component depends on x alone, through $\phi(x)$ and $r(x)$, with no dependence on (u, v) .

6.2 Decoupling bending moments

The Kirchhoff–Love closure relates the moments of Eq. (26) to the curvature mismatch,

$$\begin{aligned} M_{uu} &= D [(1 - \nu) \Delta\kappa_{uu} + \nu (\Delta\kappa_{uu} + \Delta\kappa_{vv})], \\ M_{vv} &= D [(1 - \nu) \Delta\kappa_{vv} + \nu (\Delta\kappa_{uu} + \Delta\kappa_{vv})], \\ M_{uv} &= D (1 - \nu) \Delta\kappa_{uv}, \end{aligned} \quad D := \frac{E t^3}{12 (1 - \nu^2)}. \quad (56)$$

Since Eq. (54) depends only on x , so do all three moments. This is the fact used in Section 3.4 to reduce the moment balance to the singular pair (30).

It is natural to expect the curvature mismatch to enter through the same channel as everything else — as an addition to the surface pressures $\sigma_{rr}^\pm$, which is what drives friction. It does not, and the reason is a parity argument worth spelling out.

Let the in-plane stresses carry both a membrane (mean across the thickness) and a bending

part (deviation from the mean):

$$\sigma_{ij}(\zeta) = \bar{\sigma}_{ij} + \zeta \sigma'_{ij}, \quad (57)$$

Now integrate the exact pointwise radial equilibrium equation (22) in ζ , starting from the true surface value σ_{rr}^- . Write C_0 for the membrane part of the right-hand side:

$$C_0 := -\frac{\partial \bar{\sigma}_{ur}}{\partial u} - \frac{\partial \bar{\sigma}_{vr}}{\partial v} + \kappa_{uu} \bar{\sigma}_{uu} - 2\kappa_{uv} \bar{\sigma}_{uv} + \kappa_{vv} \bar{\sigma}_{vv}, \quad (58)$$

Integrating again,

$$\sigma_{rr}(\zeta) = \sigma_{rr}^- + C_0 \left(\zeta + \frac{t}{2} \right) + \frac{6}{t^3} [\kappa_{uu} M_{uu} - 2\kappa_{uv} M_{uv} + \kappa_{vv} M_{vv}] \left(\zeta^2 - \frac{t^2}{4} \right). \quad (59)$$

The bending correction is proportional to $\zeta^2 - t^2/4$, which vanishes at *both* surfaces: at $\zeta = -t/2$ by construction of the integration constant, and at $\zeta = +t/2$ because $(t/2)^2 - t^2/4 = 0$ as well. It can bow the pressure profile in the interior of the thickness but can never shift either surface value. Hence

$$\Delta \sigma_{rr} = C_0 t = t [\kappa_{uu} \bar{\sigma}_{uu} - 2\kappa_{uv} \bar{\sigma}_{uv} + \kappa_{vv} \bar{\sigma}_{vv}] + t\mu \left[\sin 2\phi \frac{\partial \bar{\sigma}_{rr}}{\partial u} + \cos 2\phi \frac{\partial \bar{\sigma}_{rr}}{\partial v} \right], \quad (60)$$

where the second bracket comes from substituting Eq. (49) into the shear-gradient terms of C_0 , ϕ being uniform so that the derivative passes through $\sin 2\phi$ and $\cos 2\phi$. We retain this term rather than dropping it, although it is smaller than the first bracket by a factor of order t/r ; Section 8.6 identifies where it is safe to neglect.

So $\Delta \sigma_{rr}$ is unaffected by bending, and since neither surface value moves, $\bar{\sigma}_{rr}$ is unaffected too. Combined with the singularity of the moment balance found in Section 3.4, the conclusion is that in this geometry a spatially uniform curvature mismatch is invisible to the contact pressure field: the friction law (49), the recursion (51)–(52) and the transport equation (53) all stand with no bending correction. Bending does contribute to $F(x)$, but through stored energy (Section 6.3) rather than through friction.

6.3 Bending energy

The Kirchhoff–Love energy density is $u_{\text{bend}} = \frac{1}{2} M_{ij} \Delta \kappa_{ij}$, which with Eq. (56) is the standard isotropic form

$$u_{\text{bend}} = \frac{D}{2} \left[(\Delta \kappa_{uu} + \Delta \kappa_{vv})^2 - 2(1 - \nu) (\Delta \kappa_{uu} \Delta \kappa_{vv} - \Delta \kappa_{uv}^2) \right]. \quad (61)$$

Substituting the two identities of Eq. (55) collapses this to a closed form in r and ϕ alone,

$$u_{\text{bend}}(r, \phi) = \frac{D}{2} \left[\left(\frac{1}{r} - \frac{1}{r_{\text{nat}}} \right)^2 + \frac{2(1 - \nu) \sin^2 \phi}{r r_{\text{nat}}} \right], \quad (62)$$

with total stored energy, the density being uniform over the sheet,

$$U_{\text{bend}}(x) = L W u_{\text{bend}}(r(x), \phi(x)). \quad (63)$$

The two terms have distinct characters. The first is a *hoop* mismatch: it penalizes the coil radius differing from the natural radius, and vanishes when $r = r_{\text{nat}}$. The second is a *twist* mismatch, traceable to $\Delta\kappa_{uv}$ through the determinant identity; it penalizes pitch and vanishes only at $\phi = 0$. At small ϕ the twist term dominates: expanding Eq. (54), $\Delta\kappa_{uv} = O(\phi)$ whereas $\Delta\kappa_{uu}$ and $\Delta\kappa_{vv}$ are both $O(\phi^2)$. That observation reappears in Section 7.3 as the reason the initial branch of $F(x)$ is soft and linear.

7 Coupling I: the coil radius

We now combine the above governing relationships, beginning with the question: given x , what r minimizes the elastic energy in the material? In other words, what r balances the bending penalty of Eq. (62) against the stretching penalty of Eq. (64)?

7.1 The two-mode energy

The membrane energy is based on the mean strain ϵ . The long edges are traction-free, the state is uniaxial *stress* rather than uniaxial strain, so the density is $\frac{1}{2}E\epsilon^2$ per unit volume with no $1/(1-\nu^2)$ stiffening:

$$U_{\text{mem}} = \frac{1}{2} E t L W \epsilon^2. \quad (64)$$

Adding the bending energy component of Eq. (63), substituting the nondimensional variables of Eqs. (5) and factoring out the common scale, the total collapses onto a single dimensionless function of one variable:

$$U(\rho; \chi) = \frac{L W D}{2 r_{\text{nat}}^2} \tilde{U}(\rho; \chi), \quad \tilde{U} = \underbrace{\left(\frac{1}{\rho} - 1\right)^2 + \frac{2(1-\nu)\chi^2}{\rho\lambda^2}}_{\text{bending}} + \underbrace{\Gamma(\lambda-1)^2}_{\text{stretching}} \quad (65)$$

with the whole competition carried by one dimensionless group, which captures the ratio between bending and stretching stiffness:

$$\Gamma := \frac{E t r_{\text{nat}}^2}{D} = \frac{12(1-\nu^2) r_{\text{nat}}^2}{t^2} \quad (66)$$

Observe that the Young's modulus cancels out. This means the balance point between bending and stretching is therefore purely geometric, fixed by t/r_{nat} and ν alone.

Differentiating Eq. (65) at fixed χ , using $\frac{\partial \lambda}{\partial \rho} = \rho/\lambda$, the condition $\frac{\partial \tilde{U}}{\partial \rho} = 0$ simplifies to the form

$$\boxed{\underbrace{\frac{1-\rho}{\rho^3}}_{\text{hoop}} + \underbrace{\frac{(1-\nu)\chi^2(3\rho^2+\chi^2)}{\rho^2\lambda^4}}_{\text{twist}} = \underbrace{\frac{\Gamma\rho(\lambda-1)}{\lambda}}_{\text{membrane}}}. \quad (67)$$

Observe that the right-hand side vanishes only at $\lambda = 1$, where the left-hand side does not. So $\lambda > 1$ strictly for every $\chi > 0$. In other words, **the ribbon is always slightly stretched**, and “inextensible” is a limit that is never exactly attained. This residual strain, small as it is, is the only non-homogeneous source of stress to tie the friction field to a definite magnitude (Section 8.4).

There exists a unique root $\rho(\chi)$ of this equation for every $\chi \in (0, 1)$ that minimizes the total elastic energy. However, that root has no useful closed algebraic form. Clearing the denominators of Eq. (67) and squaring away the radical $\lambda = \sqrt{\rho^2 + \chi^2}$ leaves a polynomial of degree 16 in ρ , which can be solved numerically but is neither practical nor informative to solve analytically.

However, an approximation of the energy minimizing ρ can be obtained by solving for $1/\Gamma$ in the limit of large Γ —that is, $t \ll r_{\text{nat}}$. The argument takes three steps.

Step 1: the $\Gamma \rightarrow \infty$ limit. Rearrange Eq. (67) so that the only factor carrying Γ stands alone:

$$\frac{\lambda-1}{\lambda} = \frac{1}{\Gamma\rho} \left[\frac{1-\rho}{\rho^3} + \frac{(1-\nu)\chi^2(3\rho^2+\chi^2)}{\rho^2\lambda^4} \right]. \quad (68)$$

For any $\rho \in (0, 1]$ the bracketed expression is finite, so the entire right-hand side is of order Γ^{-1} , and therefore

$$\lambda - 1 = O(\Gamma^{-1}) \quad \implies \quad \lambda \rightarrow 1 \quad \text{as} \quad \Gamma \rightarrow \infty. \quad (69)$$

Now recall that $\lambda^2 = \rho^2 + \chi^2$ by Eq. (5), so imposing $\lambda = 1$ forces $\rho^2 + \chi^2 = 1$. Call the resulting radius

$$\rho_0 := \sqrt{1 - \chi^2} \quad (70)$$

so that ρ_0 is, by construction, the root of Eq. (67) in the limit of large Γ . Essentially, as $t \rightarrow 0$ and $\Gamma \rightarrow \infty$, bending becomes infinitely cheaper than stretching, and the sheet approaches inextensibility.

Step 2: expand both sides about that limit. Let Γ be large but finite, and write $\lambda = 1 + \epsilon$ with $\epsilon = O(\Gamma^{-1})$ from Eq. (69). On the right-hand side of Eq. (67),

$$\frac{\Gamma\rho(\lambda-1)}{\lambda} = \frac{\Gamma\rho\epsilon}{1+\epsilon} = \Gamma\rho\epsilon[1 - \epsilon + \dots] = \Gamma\rho_0\epsilon + O(\Gamma^{-1}), \quad (71)$$

where the two corrections absorbed into $O(\Gamma^{-1})$ are $\Gamma\rho\epsilon^2$, which is $O(\Gamma^{-1})$ because $\epsilon^2 = O(\Gamma^{-2})$, and the replacement of ρ by ρ_0 , which by Eq. (74) below costs $\Gamma\epsilon \cdot O(\Gamma^{-1})$, also $O(\Gamma^{-1})$.

On the left-hand side, the two bending terms form a smooth function of ρ and λ near $(\rho_0, 1)$, so they may simply be evaluated there, again incurring an error $O(\Gamma^{-1})$. Setting $\lambda = 1$ makes

$\lambda^4 = 1$, and $\chi^2 = 1 - \rho_0^2$ eliminates χ ; the latter also collapses the twist numerator, since $3\rho_0^2 + \chi^2 = 3\rho_0^2 + 1 - \rho_0^2 = 1 + 2\rho_0^2$. The left-hand side is therefore

$$\underbrace{\frac{1 - \rho_0}{\rho_0^3}}_{\text{hoop}} + \underbrace{\frac{(1 - \nu)(1 - \rho_0^2)(1 + 2\rho_0^2)}{\rho_0^2}}_{\text{twist}}. \quad (72)$$

Step 3: solve for the strain. Equating Eqs. (71) and (72) and dividing through by $\Gamma\rho_0$,

$$\boxed{\epsilon(\chi) = \frac{(1 - \rho_0) + (1 - \nu)\rho_0(1 - \rho_0^2)(1 + 2\rho_0^2)}{\Gamma\rho_0^4} + O(\Gamma^{-2})} \quad (73)$$

where the common factor ρ_0^{-4} has been taken outside. Finally, the radius follows:

$$\rho = \sqrt{(1 + \epsilon)^2 - \chi^2} \quad \implies \quad \rho(\chi) = \sqrt{\rho_0^2 + 2\epsilon(\chi)}, \quad (74)$$

by dropping the $O(\Gamma^{-2})$ order ϵ^2 term once more.

The relative error in ϵ only exceeds 1% after $x/L \approx 0.95$. So the closed form is accurate to a fraction of a percent over essentially the whole pull, degrading only in the final percent of extension—which Section 7.2 will show is physically unreachable anyways. Therefore, for practical purposes, we will use Eq. (73) for the energy minimizing strain and radius in the remainder of the paper.

Panel (a) also displays the feature of $\epsilon(\chi)$ most worth internalizing. The strain is of order 10^{-7} at small extension and climbs through four orders of magnitude, reaching a few times 10^{-3} only as the coil stops tightening. The ribbon is *barely* stretched over almost the whole pull — but it is never exactly unstretched, and Section 8.4 shows that this is the entire origin of the friction force.

7.2 The sliding-to-stretching crossover

Solving Eq. (67) numerically over the whole range shows ρ decreasing *monotonically* in χ , with no turning point: $\frac{d\rho}{dx} < 0$ everywhere. Pushed far past $x = L$ —which the ribbon permits with a large enough ϵ —the balance in Eq. (67) between the hoop term ρ^{-3} and the membrane term settles on

$$\rho \approx \left[\frac{\chi}{\Gamma(\chi - 1)} \right]^{1/4} \xrightarrow{\chi \gg 1} \rho \rightarrow \Gamma^{-1/4} \quad (75)$$

The radius monotonically approaches a small value asymptote. What exists is therefore not a floor but a change of regime.

Differentiate $\lambda^2 = \rho^2 + \chi^2$ (Eq. 5) with respect to χ :

$$\boxed{\underbrace{\lambda \frac{d\lambda}{d\chi}}_{\text{into stretching}} - \underbrace{\rho \frac{d\rho}{d\chi}}_{\text{into tightening}} = \chi} \quad (76)$$

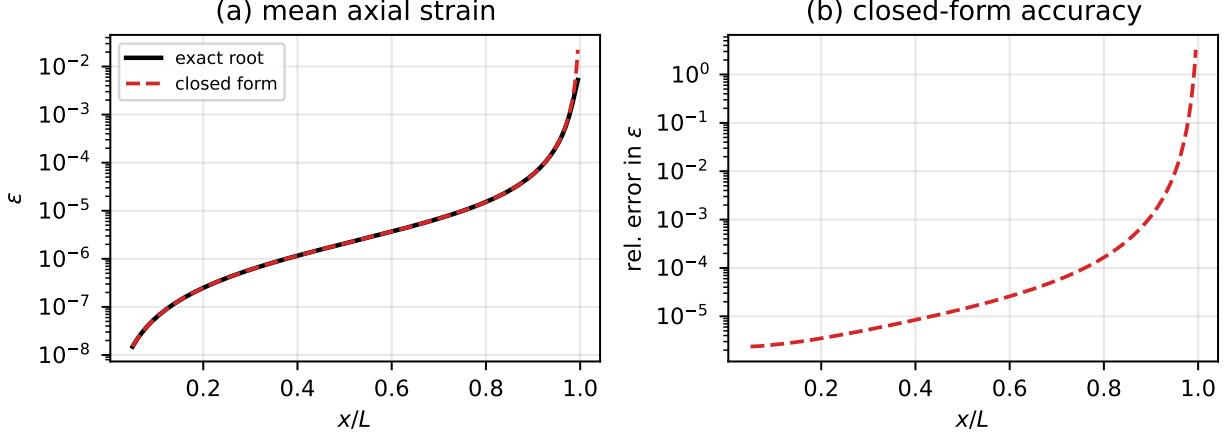


Figure 2: (a) The mean axial strain: the closed form Eq. (73) against the exact root of Eq. (67), on a logarithmic scale. The strain is tiny but strictly positive everywhere, and rises steeply only near the crossover x_* of Eq. (78). (b) Relative error of the closed form, with the 1% level marked; it is accurate until the expansion’s own validity limit. (c) The capstan gain $\mathcal{A}$ of Eq. (98) measured from the two-dimensional solve, against the accumulation exponent $\lambda_+ L$ of Eq. (105) that drives it.

In other words, an increment of pull is spent partly on lengthening the ribbon (λ grows) and partly on tightening the coil (ρ shrinks, so the second term is positive), and the two shares always sum to χ . Both shares are therefore well defined at every extension, with no expansion and no threshold. Define the crossover $*$ as the extension at which they are *equal*, each taking half of the increment:

$$\lambda \left. \frac{d\lambda}{d\chi} \right|_* = -\rho \left. \frac{d\rho}{d\chi} \right|_* = \frac{\chi_*}{2}. \quad (77)$$

Before χ_* , the pull is absorbed almost entirely by the material sliding and bending around itself. After it, it is almost entirely stretching. Because most materials do not survive stretching at large strains, this crossover appears as an effective “locking” of the pull.

Evaluating Eq. (77) on the solution of Eq. (67) gives, to within a fraction of a percent across five decades of Γ ,

$$\boxed{\rho_* \approx \left(\frac{4}{\Gamma}\right)^{1/6}, \quad \chi_* \approx \sqrt{1 - \rho_*^2}, \quad \epsilon_* \approx \frac{\rho_*^2}{2}.} \quad (78)$$

In dimensional form the crossover radius is

$$r_* = r_{\text{nat}} \left(\frac{4}{\Gamma}\right)^{1/6} = \frac{(r_{\text{nat}}^2 t)^{1/3}}{(3(1 - \nu^2))^{1/6}} \sim (r_{\text{nat}}^2 t)^{1/3}, \quad (79)$$

and the crossover displacement is

$$x_* = L \chi_* \approx L \sqrt{1 - \left(\frac{4}{\Gamma}\right)^{1/3}} \quad (80)$$

and the crossover strain is

$$\epsilon_* \approx \frac{1}{2} \left(\frac{4}{\Gamma} \right)^{1/3} \sim \left(\frac{t}{r_{\text{nat}}} \right)^{2/3}. \quad (81)$$

The below has not been reviewed

Packing is the same mechanism, not a second one. Eq. (13) gave a geometric limit ρ_{jam} at which the innermost turn closes on the axis, and it is natural to want to combine it with ρ_* . Writing $\rho_{\text{stop}} = \max(\rho_*, \rho_{\text{jam}})$ would be wrong on two counts: it treats packing as a hard wall, and it treats the two as alternatives when they should act together. Both objections have the same resolution, which is that *these are not two mechanisms*. Each is the bending energy diverging as some radius goes to zero — for ρ_* the mean radius, for ρ_{jam} the innermost one — and both can be captured by writing the bending energy over the true radius distribution instead of at the mean.

With turns stacked one thickness apart, the local radius at fractional position $\varsigma \in [-\frac{1}{2}, \frac{1}{2}]$ along the ribbon is

$$\rho_{\text{loc}}(\varsigma) = \rho + 2\rho_{\text{jam}} \varsigma, \quad (82)$$

running from $\rho - \rho_{\text{jam}}$ at the innermost turn to $\rho + \rho_{\text{jam}}$ at the outermost, which reproduces Eq. (13) exactly: the inner radius vanishes when $\rho = \rho_{\text{jam}}$. Averaging the two radius-dependent factors of Eq. (62) over ς gives closed forms,

$$\left\langle \frac{1}{\rho_{\text{loc}}^2} \right\rangle = \frac{1}{\rho^2 - \rho_{\text{jam}}^2}, \quad \left\langle \frac{1}{\rho_{\text{loc}}} \right\rangle = \frac{1}{2\rho_{\text{jam}}} \ln \left(\frac{\rho + \rho_{\text{jam}}}{\rho - \rho_{\text{jam}}} \right) =: \mathcal{L}, \quad (83)$$

and substituting them into Eq. (65) in place of ρ^{-2} and ρ^{-1} replaces that energy with

$$\boxed{\tilde{U} = \frac{1}{\rho^2 - \rho_{\text{jam}}^2} - 2\mathcal{L} + 1 + \frac{2(1-\nu)\chi^2}{\lambda^2} \mathcal{L} + \Gamma(\lambda - 1)^2.} \quad (84)$$

This is one smooth energy with no max anywhere in it. As $\rho_{\text{jam}} \rightarrow 0$, $\mathcal{L} \rightarrow 1/\rho$ and Eq. (84) reduces term by term to Eq. (65). As $\rho \rightarrow \rho_{\text{jam}}$ both $(\rho^2 - \rho_{\text{jam}}^2)^{-1}$ and $\mathcal{L}$ diverge, so the packing limit appears as an *asymptote* of the same bending energy rather than as an imposed constraint — the coil approaches it and never reaches it, exactly as it approaches Eq. (75) in the thin-stack limit.

The two contributions genuinely add. Minimizing Eq. (84) numerically at the parameters of Section 9.2, the radius attained near $x \approx L$ exceeds *both* ρ_* and ρ_{jam} through the region where they are comparable, by up to about 15% — a blend, not a switch. Which one sets the scale is a property of the winding alone. Since $\rho_{\text{jam}} = \frac{1}{2} n t / r_{\text{nat}}$ grows linearly in the number of turns while $\rho_* \propto (t/r_{\text{nat}})^{1/3}$ does not, their ratio is

$$\frac{\rho_{\text{jam}}}{\rho_*} = \frac{n}{2} \left(\frac{t}{r_{\text{nat}}} \right)^{2/3} (3(1-\nu^2))^{1/6}, \quad (85)$$

so packing dominates for many turns or a thick ribbon, and the elastic mechanism dominates for few turns or a thin one, with the boundary $n \propto (t/r_{\text{nat}})^{-2/3}$. Figure 3(c) maps this out. The coil of

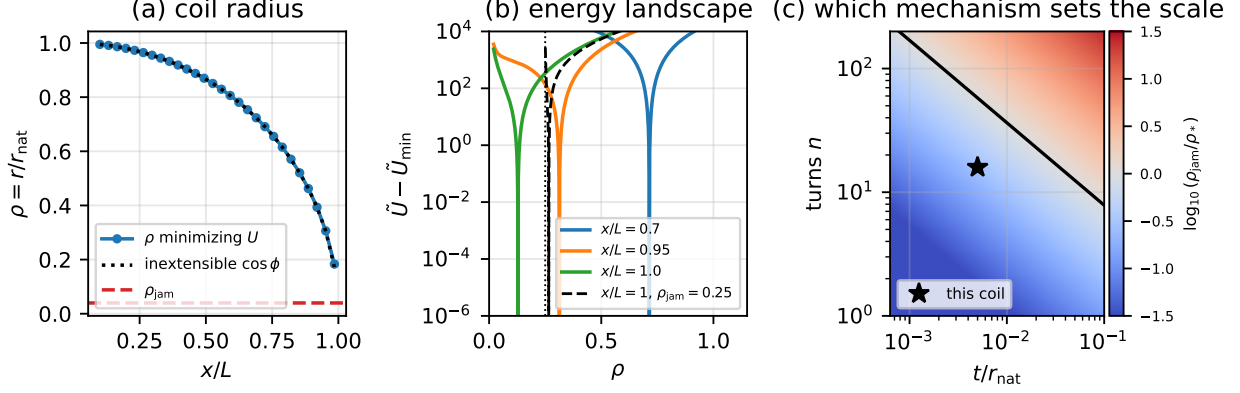


Figure 3: (a) The energy-minimizing radius from Eq. (67), against the inextensible $\cos \phi$ and the packing floor Eq. (13). The minimizer tracks the inextensible value closely, consistent with $\Gamma \gg 1$, and decreases monotonically throughout — there is no turning point. (b) The energy landscape of Eq. (84) at three extensions with $\rho_{\text{jam}} = 0$, plus one case with $\rho_{\text{jam}} = 0.25$ (dashed) showing the packing divergence appear as an asymptote of the same energy. The well is single and deep, so the minimization is unambiguous. (c) Which mechanism sets the radius scale, from Eq. (85): the black contour is $\rho_{\text{jam}} = \rho_*$, and the star marks the coil of Section 9.2.

Section 9.2 sits at $\rho_{\text{jam}}/\rho_* = 0.28$, well inside the elastically dominated region, which is why ρ_{jam} appears to do so little there; a coil wound with ten times as many turns would sit on the other side of the line. Because that ratio is 0.28, the numerical work of Section 9 uses the $\rho_{\text{jam}} \rightarrow 0$ limit Eq. (65) rather than Eq. (84); the correction factor $\rho^2 / (\rho^2 - \rho_{\text{jam}}^2)$ stays below 1.05 over the solved range.

What the crossover means for the experiment. Past x_* the pull goes mostly into strain, so the strain climbs steeply while the coil radius nearly stalls. This is the model’s account of lock-up, and it is a crossover rather than an event: nothing is pinned, nothing is discontinuous, and the coil is still tightening slowly on the far side of it. Whether a given sheet survives to display the change is a separate question — ϵ_* from Eq. (78) must lie below the material’s failure strain — and for paper the two are comparable, which is why paper tears rather than stretching visibly.

7.3 The conjugate force

With $\rho(x)$ determined, the force follows without further work, and with a simplification worth pointing out. Along the minimizing path,

$$F(x) = \frac{d}{dx} U(\rho(x); x) = \left. \frac{\partial U}{\partial x} \right|_{\rho} + \underbrace{\frac{\partial U}{\partial \rho} \frac{d\rho}{dx}}_{=0} = \left. \frac{\partial U}{\partial x} \right|_{\rho}, \quad (86)$$

the second term vanishing because $\frac{\partial U}{\partial \rho} = 0$ is exactly the stationarity condition (67). This is the envelope theorem, and the word “stationary” is doing specific work here: U is stationary against variations of ρ at fixed x , which is what kills the second term, while the root $\rho(x)$ itself certainly

does vary with x — indeed $\frac{d\rho}{dx} \neq 0$, as Figure 3(a) shows. The content of Eq. (86) is that we never need its value. So $\frac{d\rho}{dx}$ is never needed: the work $F dx$ is the *partial* derivative at frozen radius, evaluated at the minimizer. One consequence is not obvious in advance — the hoop term $(1/\rho - 1)^2$ of Eq. (65) carries no explicit x , so it contributes *nothing* directly to the force, despite being the term that diverges as the coil tightens. It acts only indirectly, by setting where $\rho(x)$ sits. Differentiating the remaining two terms,

$$F(x) = \underbrace{\frac{2(1-\nu)WD}{r_{\text{nat}}^2} \frac{\rho\chi}{\lambda^4}}_{\text{twist (bending)}} + \underbrace{EtW\epsilon \frac{\chi}{\lambda}}_{\text{membrane tension}}, \quad (87)$$

the second term following from $\Gamma WD/r_{\text{nat}}^2 = EtW$ by Eq. (66). It is exactly the ribbon's axial tension $EtW\epsilon$ projected through $\chi/\lambda = \sin\phi$, as it must be. Both terms are of the same order in general: ϵ is small, of order $1/\Gamma$, but it is multiplied by the large Et .

Eq. (87) can be made independent of Γ altogether. Substituting $\Gamma(\lambda - 1)/\lambda$ from the stationarity condition (67) — an exact rearrangement, not an approximation —

$$F(x) = \frac{WD}{r_{\text{nat}}^2} \left[\frac{\chi(1-\rho)}{\rho^4} + \frac{(1-\nu)\chi}{\lambda^4} \left(2\rho + \frac{\chi^2(3\rho^2 + \chi^2)}{\rho^3} \right) \right]. \quad (88)$$

Once $\rho(x)$ is known the force does not depend on the membrane stiffness at all; Γ enters only through how far it lets the coil tighten. Since $\rho > \rho_0$ always and the leading bracket term falls steeply with ρ , the extensible force is always *below* the inextensible one — stretching is a relief mechanism, as expected.

Setting $\lambda \rightarrow 1$, $\rho \rightarrow \cos\phi$ and $\chi \rightarrow \sin\phi$ gives the inextensible closed form, which is the practical one:

$$F_{\text{bend}}(x) = \frac{WD}{r_{\text{nat}}^2} \left[\frac{\sin\phi(1-\cos\phi)}{\cos^4\phi} + (1-\nu) \frac{2\sin\phi\cos^2\phi + \sin^3\phi}{\cos^3\phi} \right], \quad \phi = \arcsin \frac{x}{L}. \quad (89)$$

This agrees with Eq. (88) identically, via the trigonometric identity $2\cos^4\phi + 3\sin^2\phi\cos^2\phi + \sin^4\phi = 2\cos^2\phi + \sin^2\phi$, which is a useful independent check on the whole envelope argument. Its two limits are

$$F_{\text{bend}} \approx \frac{2(1-\nu)WD}{r_{\text{nat}}^2} \frac{x}{L} \quad (x \ll L), \quad F_{\text{bend}} \approx \frac{WD}{4r_{\text{nat}}^2(1-x/L)^2} \quad (x \rightarrow L). \quad (90)$$

The first is linear and soft, and — per the remark closing Section 6.3 — is driven entirely by the twist mismatch, the hoop term contributing only at $O(\chi^4)$. The second diverges as $(1-x/L)^{-2}$, steeply enough to pass for an exponential over any limited fitting window.

8 Coupling II: the friction subsystem

8.1 Assembling the equations

Substituting the friction law (49)–(50) into the thickness-averaged in-plane equilibrium equations (23)–(24), and simplifying the curvature terms with $\kappa_{uu} \sin 2\phi - \kappa_{uv} \cos 2\phi = \kappa_{uv}$ (a double-angle identity applied to Eq. 10), gives the friction body force of Eq. (36) explicitly:

$$\boxed{f_u^b = \mu \sin 2\phi \left[\frac{\Delta\sigma_{rr}}{t} + \frac{\bar{\sigma}_{rr}}{2r} \right], \quad f_v^b = \mu \cos 2\phi \left[\frac{\Delta\sigma_{rr}}{t} - \frac{\bar{\sigma}_{rr}}{2r} \right].} \quad (91)$$

Note that $\Delta\sigma_{rr}/t$ contains no small denominator in practice: by Eq. (60) it equals $\kappa_{uu}\bar{\sigma}_{uu} - 2\kappa_{uv}\bar{\sigma}_{uv} + \kappa_{vv}\bar{\sigma}_{vv}$ plus an $O(t)$ correction.

Counting equations and unknowns. There are five unknown fields of (u, v) : $\bar{\sigma}_{uu}$, $\bar{\sigma}_{uv}$, $\bar{\sigma}_{vv}$ and σ_{rr}^+ , σ_{rr}^- (equivalently $\bar{\sigma}_{rr}$ and $\Delta\sigma_{rr}$). It is tempting to count every relation derived so far and declare closure once the totals match, but that overstates what each buys. The radial equilibrium equation (60) and the contact recursion (53) are both statements about $\Delta\sigma_{rr}$ in terms of $\bar{\sigma}_{rr}$; together they are exactly two equations for the two quantities $\bar{\sigma}_{rr}$ and $\Delta\sigma_{rr}$, and once the in-plane stresses are known that pair is self-contained. It supplies the in-plane problem with a source term but no additional constraint. So the real gap is the one plane-stress elasticity always has: the two in-plane equilibrium equations (36) relate three unknowns, leaving one relation missing. Compatibility (34) is that relation. Properly accounted, the closed system is

two in-plane equilibrium + one compatibility	for $\bar{\sigma}_{uu}, \bar{\sigma}_{uv}, \bar{\sigma}_{vv}$
one radial equilibrium + one contact recursion	for $\sigma_{rr}^+, \sigma_{rr}^-$

— five equations for five unknowns, but arranged as two coupled subsystems rather than five constraints on everything at once.

Boundary conditions. The long edges are traction-free, so $\bar{\sigma}_{uv} = \bar{\sigma}_{vv} = 0$ at $v = \pm W/2$. Since $\bar{\sigma}_{vr}$ is tied directly to $\bar{\sigma}_{rr}$ by Eq. (49), a traction-free edge also forces $\sigma_{rr}^\pm = 0$ there individually, hence $\bar{\sigma}_{rr} = \Delta\sigma_{rr} = 0$ at the edges. Note that $\bar{\sigma}_{uu}$ is *not* constrained at a free v -edge: the traction on a face with normal $\hat{v}$ involves σ_{vv} , σ_{uv} and σ_{vr} only. At the active end, $\sigma_{rr}^+ = 0$ at $u = L$, already supplied geometrically by Section 5.4. The resultant transmitted through the cross-section at $u = L$, projected onto the pulling direction, is the applied force:

$$\boxed{F(x) = W t [\bar{\sigma}_{uu}(L) \sin \phi + \bar{\sigma}_{uv}(L) \cos \phi],} \quad (92)$$

the factor Wt being the cross-sectional area and the bracket the traction on a face of normal $\hat{u}$ resolved along z , using $\hat{u} \cdot \hat{z} = \sin \phi$ and $\hat{v} \cdot \hat{z} = \cos \phi$ from Eq. (9). The opposite reaction force is applied at the other end, $u = 0$.

8.2 The Beltrami–Michell form

Eq. (34) still contains a mixed derivative of $\bar{\sigma}_{uv}$, which we would rather not carry. The two in-plane equilibrium equations (36) already relate the first derivatives of $\bar{\sigma}_{uv}$ to the other fields: differentiating the first with respect to u , the second with respect to v , and adding,

$$2 \frac{\partial^2 \bar{\sigma}_{uv}}{\partial u \partial v} = - \left(\frac{\partial^2 \bar{\sigma}_{uu}}{\partial u^2} + \frac{\partial^2 \bar{\sigma}_{vv}}{\partial v^2} \right) + \frac{\partial f_u^b}{\partial u} + \frac{\partial f_v^b}{\partial v}. \quad (93)$$

Substituting into Eq. (34), the coefficients of $\frac{\partial^2 \bar{\sigma}_{uu}}{\partial u^2}$ and $\frac{\partial^2 \bar{\sigma}_{vv}}{\partial v^2}$ both collect to $-\nu + (1 + \nu) = 1$, matching the coefficient already on the other two second derivatives, so the four combine into a plain Laplacian:

$$\boxed{\nabla^2 (\bar{\sigma}_{uu} + \bar{\sigma}_{vv}) = (1 + \nu) \left(\frac{\partial f_u^b}{\partial u} + \frac{\partial f_v^b}{\partial v} \right)}. \quad (94)$$

Intuitively this is a Poisson equation: the total in-plane stress cannot vary however it likes, but must curve in response to how unevenly the friction forcing is distributed, the way a temperature field curves in response to a heat source.

Eq. (94) is worth recognizing: it is exactly the standard plane-stress Beltrami–Michell equation $\nabla^2 \sigma_{kk} = -(1 + \nu) \frac{\partial b_k}{\partial x_k}$ for a body force $b_i = -f_i^b$. That identification is the promised verification that the displacement formulation of Section 4.3 solves the same problem: Eqs. (37)–(38) and Eqs. (36), (94) are two presentations of one system, and the displacement form is preferable only because it satisfies compatibility identically instead of imposing it as an extra second-order equation.

8.3 Normalization: what sets the magnitude

Every equation of this subsystem is linear and *homogeneous*. The friction body force (91) is proportional to the contact pressure, the contact pressure is generated by the in-plane stress through Eq. (60), and the traction-free conditions are homogeneous too. The trivial solution — zero stress everywhere — therefore satisfies the entire subsystem at every x . Something outside it must fix the scale, and it is worth being precise about what.

The resolution is the arc-length constraint, Eq. (2), which so far has been used only to determine ρ . It states that the ribbon’s material length and its length in space must agree, i.e. that the mean axial strain is not free but equal to $\lambda - 1$:

$$\boxed{\frac{1}{LW} \int_0^L \int_{-W/2}^{W/2} \bar{\epsilon}_{uu} \, dv \, du = \frac{1}{LW} \int_0^L \int_{-W/2}^{W/2} \frac{\bar{\sigma}_{uu} - \nu \bar{\sigma}_{vv}}{E} \, dv \, du = \lambda(\rho(x)) - 1}. \quad (95)$$

This single scalar relation is inhomogeneous — its right-hand side is nonzero for every $x > 0$, by the strict inequality $\lambda > 1$ established in Section ?? — so it excludes the trivial solution and fixes the amplitude. In the displacement formulation it takes the especially clean form of a Dirichlet condition on the ends,

$$a(L, v) - a(0, v) = L(\lambda - 1), \quad (96)$$

which is how Section 9 imposes it. The picture is that the governing equations determine the *shape*

of the stress field, and the ribbon’s own inextensibility — almost, but not quite, exact — determines its *amplitude*.

Two honest caveats. First, this ties the friction field’s magnitude to the geometric stretch, and those are not obviously the same physical quantity: a self-equilibrated friction pre-stress and an imposed elongation are different things, and Eq. (95) couples them. Second, the amplitude so obtained is small, since $\lambda - 1 = O(1/\Gamma)$, and it must then be multiplied by whatever amplification the friction loop provides (Section 8.6) — a product of a very small number and a very large one, which is numerically delicate. Section 9.3 quantifies how delicate.

8.4 From the residual strain to the measured force

The residual strain established in Section ?? is what seeds the friction field. It is worth following that chain explicitly, because doing so also settles what “the” force is.

There is one measurable force: the resultant Eq. (92) transmitted through the cross-section the Instron grips. Eq. (87) is not a rival value for it but a *decomposition* of it, and the friction subsystem modifies exactly one of the two terms. To see this, read the second term of Eq. (87) again: it is $EtW\epsilon\chi/\lambda$, which is precisely Eq. (92) evaluated on a *uniform* strain field. With $\bar{\sigma}_{uu} = E\epsilon$ everywhere and $\bar{\sigma}_{uv} = 0$,

$$Wt\bar{\sigma}_{uu}(L)\sin\phi = WtE\epsilon\frac{\chi}{\lambda} \quad (97)$$

identically. Friction does not add a new term to the force; it makes $\bar{\sigma}_{uu}$ non-uniform, concentrating it towards the ends, so that its value at $u = L$ exceeds the uniform-strain value. Naming that ratio the capstan gain,

$$\mathcal{A}(x) := \frac{\bar{\sigma}_{uu}(L)}{E\epsilon(x)}, \quad \mathcal{A} \rightarrow 1 \text{ as } \mu \rightarrow 0, \quad (98)$$

the entire force law collapses into a single expression:

$$F(x) = \underbrace{\frac{2(1-\nu)WD}{r_{\text{nat}}^2} \frac{\rho\chi}{\lambda^4}}_{\text{twist, friction-independent}} + \underbrace{\mathcal{A}(x)EtW\epsilon(x)\frac{\chi}{\lambda}}_{\text{membrane, capstan-amplified}}. \quad (99)$$

Setting $\mathcal{A} = 1$ recovers Eq. (87) exactly, and hence Eq. (89) in the inextensible limit. The twist term is untouched by friction, for two reasons already established: bending does not couple to the contact pressure (Section 6.2), and $\rho(x)$ is fixed by the friction-free balance of Section ??.

The causal chain from geometry to force is therefore

$$\underbrace{\lambda > 1}_{\text{Eq. (67)}} \longrightarrow \underbrace{\epsilon = \frac{H(\rho_0)}{\Gamma\rho_0^4}}_{\text{Eq. (73)}} \longrightarrow \underbrace{E\epsilon}_{\text{uniform membrane stress}} \longrightarrow \underbrace{\mathcal{A}E\epsilon}_{\text{friction amplification}} \longrightarrow \underbrace{F}_{\text{Eq. (99)}}, \quad (100)$$

and no step in it is adjustable. Every arrow is an equation already derived. The first arrow is the one that carries the weight: were $\lambda = 1$ exactly, every subsequent quantity would vanish and the model would predict no friction force at all. The strict inequality $\lambda > 1$ of Section ?? is what makes

the friction branch exist.

The gain itself is measured directly from the two-dimensional solution (Section 9.2) and is plotted in Figure 2(c). It runs from about 2 to about 100, peaking near $x/L \approx 0.45$, and tracks the accumulation exponent $\lambda_+ L$ of Eq. (105) as it should — the gain is what that exponent buys once the boundary conditions have taken their cut. It is also why the two terms of Eq. (99) trade places: the twist term dominates below $x/L \approx 0.2$, and the amplified membrane term above it.

8.5 Why this is not an energy minimization

It is tempting to treat the whole coupled problem the way Section 7 treated the radius: write the total stored energy including the friction-driven membrane stress, and minimize. This is wrong, and the reason is worth recording because the error is not obvious.

Minimization is legitimate only for a recoverable potential. The friction-amplified membrane stress does store elastic energy, but that energy is held in place by a dissipative, history-dependent contact law: release the load and the stress does not simply run back down the same path. In a rate-independent quasi-static model there is no way to separate the stored-and-recoverable part from the stored-but-friction-locked part, so the total membrane energy is not an admissible potential for a variational argument.

The consequence is quantitative rather than merely philosophical. Feeding the solved field's strain non-uniformity $\Xi := \langle \bar{\epsilon}_{uu}^2 \rangle / \langle \bar{\epsilon}_{uu} \rangle^2$ back into Eq. (65) as an effective membrane stiffening gives $\Xi \sim 4 \times 10^5$ for the parameters of Section 9.2, which drives the minimizer of Eq. (67) to the inextensible radius, collapses ϵ to about 10^{-12} , and collapses F with it. Only the bare, uniform-strain membrane stiffness of Eq. (64) belongs in the radius balance.

The same objection disqualifies a tempting cross-check. One might try to obtain F a second way, as $\frac{dU_{\text{elastic}}}{dx}$ plus a Coulomb dissipation integral, and compare it against Eq. (99). That is not an independent check but an inadmissible one: it requires splitting the stored energy into recoverable and friction-locked parts, which is precisely what a rate-independent model cannot do. Carried out anyway, it disagrees with Eq. (99) by factors of two to five — a symptom of the illegitimacy, not evidence of an error in either. Eq. (99), one force decomposed into two additive terms, is the statement to use.

8.6 Width-averaged reduction and the accumulation exponent

Before solving the full two-dimensional system it is worth extracting the essential behaviour analytically, because it explains both the shape of the answer and the numerical difficulty.

Integrate the in-plane $\hat{u}$ equation (36) across the width and divide by W . The $\frac{\partial \bar{\sigma}_{uv}}{\partial v}$ term integrates exactly to its edge values, which vanish, so with $\hat{\sigma}_{uu}(u)$ and $\hat{\sigma}_{rr}(u)$ denoting the width averages of $\bar{\sigma}_{uu}$ and $\bar{\sigma}_{rr}$, and using Eq. (60) at leading order for $\Delta \sigma_{rr}$,

$$\frac{d\hat{\sigma}_{uu}}{du} = \mu \sin 2\phi \left[\kappa_{uu} \hat{\sigma}_{uu} + \frac{\hat{\sigma}_{rr}}{2r} \right]. \quad (101)$$

Doing the same to the transport equation (53), whose $\frac{\partial \bar{\sigma}_{rr}}{\partial v}$ term likewise integrates to vanishing

edge values,

$$\frac{d\hat{\sigma}_{rr}}{du} = \frac{t \kappa_{uu}}{2\pi r \cos \phi} \hat{\sigma}_{uu}. \quad (102)$$

Both steps are exact consequences of the edge conditions rather than mode truncations. This is a linear, constant-coefficient 2×2 system in u , the coefficients depending on x only: $\frac{d}{du} (\hat{\sigma}_{uu}, \hat{\sigma}_{rr})^\top = \begin{pmatrix} A & B \\ C & 0 \end{pmatrix} (\hat{\sigma}_{uu}, \hat{\sigma}_{rr})^\top$ with

$$A = \mu \sin 2\phi \kappa_{uu}, \quad B = \frac{\mu \sin 2\phi}{2r}, \quad C = \frac{t \kappa_{uu}}{2\pi r \cos \phi}. \quad (103)$$

Its trace is A and its determinant $-BC < 0$, so the eigenvalues

$$\lambda_{\pm} = \frac{A \pm \sqrt{A^2 + 4BC}}{2} \quad (104)$$

are always *real*, with $\lambda_+ > 0 > \lambda_-$. The system is a saddle, never a rotation, so $\hat{\sigma}_{uu}$ and $\hat{\sigma}_{rr}$ are sums of real exponentials in u and never oscillatory — a useful check, since a mode-truncated treatment of the same equations can produce spurious oscillation by coupling a mode to a frequency the edge conditions do not admit. Since $C \propto t$ is small, $\lambda_+ \approx A$ and the growing mode is

$$\boxed{\lambda_+ \approx \mu \sin 2\phi \kappa_{uu} = \frac{\mu \sin 2\phi \cos^2 \phi}{r} \implies \lambda_+ L \approx 0.77 \mu \theta_L.} \quad (105)$$

The last step uses $r = r_{\text{nat}} \cos \phi$ and $L/r_{\text{nat}} = \theta_L$, then maximizes $\sin 2\phi \cos \phi \leq 0.77$ over ϕ . This is the single most consequential number in the friction subsystem, so it is worth reading carefully: the total amplification exponent along the ribbon is set by the friction coefficient times the *number of turns*, and is completely independent of E , t and W . For $\mu = 0.3$ and thirty turns it is 46, an amplification of $e^{46} \sim 10^{20}$ between the two ends of the ribbon. This is the capstan effect, and it is why the friction contribution is at once physically dominant and numerically treacherous.

9 Numerical solution

9.1 Formulation

The full two-dimensional system is solved on a finite-difference grid in (u, v) at each frozen x . Four choices are worth recording, each made to avoid a specific failure.

1. **Displacement unknowns.** The primary fields are a and b of Eq. (35), solved through Eqs. (37)–(38), rather than the three stresses through Eqs. (36) and (94). Compatibility is then identical rather than imposed, which makes the assembled system square and consistent; and the friction body force appears undifferentiated, so nothing interpolated is ever differentiated.
2. **Exact contact recursion.** σ_{rr}^+ is carried as a third unknown field satisfying Eqs. (51)–(52) directly, with bilinear interpolation at the generally off-grid contact partner, rather than

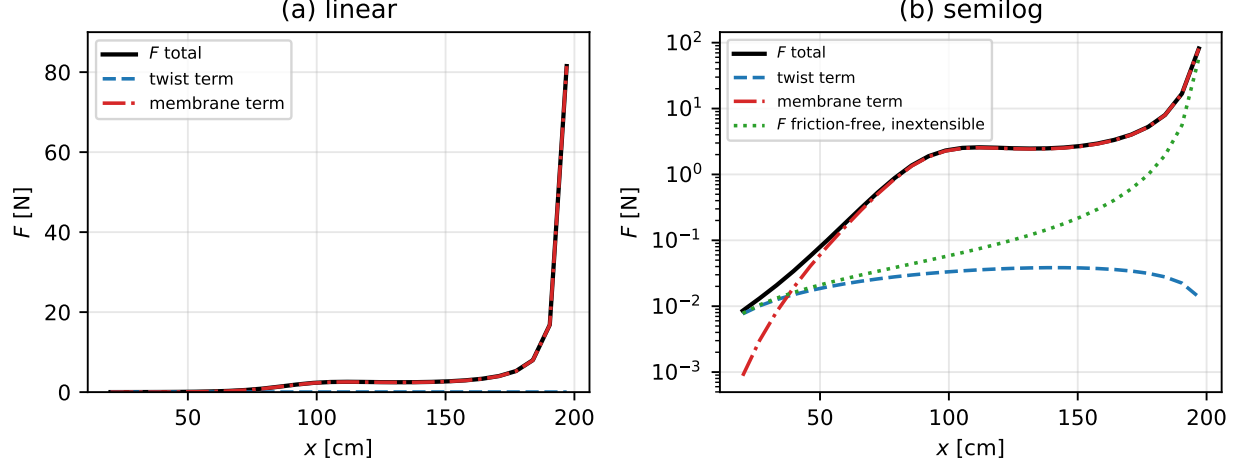


Figure 4: Force–displacement, linear and semilogarithmic, showing the two additive terms of Eq. (99). The twist term is friction-independent and dominates below $x/L \approx 0.2$; the capstan-amplified membrane term dominates above it. The dotted curve is the friction-free closed form Eq. (89) for comparison, which the total exceeds once the gain $\mathcal{A}$ of Eq. (98) departs from unity. The total spans 0.0085 to 82 N with a logarithmic slope that *increases* with x , i.e. growth faster than any single exponential.

the Taylor-expanded transport equation (53). This enforces $\sigma_{rr}^+(L, v) = 0$ geometrically and avoids finite-differencing a first-order equation across the narrow bands where both neighbours are present.

3. **Elongation as a Dirichlet condition.** Eq. (96) supplies the normalization, so no arbitrary scale anchor is needed anywhere. Laterally, $u = 0$ is fully clamped ($b = 0$) and $u = L$ is left shear-free ($\bar{\sigma}_{uv} = 0$). Clamping both ends laterally places a clamped-meets-free corner singularity at $u = L$, exactly where Eq. (92) is evaluated; imposing $\bar{\sigma}_{uv} = 0$ at both ends instead over-constrains the global axial balance, because the equilibrium equations are written in the rotating local frame, so $\iint f_u^b du dv$ is not a conserved quantity.
4. **Resolution set by turns, not by length.** The contact partner sits $2\pi r \cos \phi$ away in u , so the recursion is meaningless unless several grid points fit inside that distance. Measured convergence: F is converged to a few percent at ten or more points per turn, and not converged at all below three. Since the offset shrinks as the coil tightens, the grid is adapted per x , reaching several thousand points along u near full extension.

9.2 Results

Figures 3–5 use $L = 200$ cm, $W = 8$ cm, $t = 0.01$ cm, $r_{\text{nat}} = 2$ cm (hence $\theta_L = 100$ rad, about 15.9 turns), $E = 3 \times 10^5$ N/cm², $\nu = 0.3$ and $\mu = 0.3$. These give $\Gamma = 4.4 \times 10^5$, $\rho_* = 0.13$ and $\rho_{\text{jam}} = 0.040$ — so the elastic floor of Eq. (78) is the operative one — and a peak amplification exponent $\lambda_+ L \approx 23$.

Three results stand out.

First, F_{bend} is strictly monotonic and grows faster than exponentially: its logarithmic slope $\frac{d(\ln F)}{dx}$ rises from 0.020 to 0.36 cm^{-1} across the sweep. It spans four orders of magnitude, 0.0078 to 58 N, with no fitted parameter anywhere. The closed form (89) and the numerically differentiated energy agree to within a couple of percent (Figure 4), which validates the envelope argument of Section 7.3 and the numerical energy bookkeeping against each other.

Second, the total stored elastic energy is overwhelmingly bending, not membrane: $U_{\text{mem}}/U_{\text{bend}}$ runs from 10^{-4} to 10^{-3} over almost the whole range, rising to 0.4 only in the last few percent of extension. So $\frac{dU_{\text{elastic}}}{dx}$ is, to a fraction of a percent, the pure bending force of Eq. (89). The elastic branch of the model is essentially entirely Section 6’s physics.

Third, the capstan-amplified membrane term of Eq. (99) overtakes the twist term at $x/L \approx 0.2$ and dominates thereafter, the measured gain $\mathcal{A}$ of Eq. (98) reaching about 100 near $x/L \approx 0.45$ (Figure 2c). The total force shows a shallow non-monotonic plateau between $x/L \approx 0.53$ and 0.72. That plateau is not numerical noise: it tracks the contact-area collapse predicted in Section 5.1, the fraction of the sheet possessing an inner neighbour falling from 0.79 to 0.22 and recovering to 0.69 across the sweep as the axial offset $2\pi r \sin \phi$ passes through its maximum. Its depth is governed by $W/(\pi r_{\text{nat}})$, which is 1.27 here; at 2.04 the contact fraction only falls to 0.52 and the plateau largely disappears.

9.3 Limits of the numerical treatment

These are stated in full because they bound what the friction half of the model can currently be said to predict.

A conditioning wall. By Eq. (105) the amplification exponent is $\lambda_+ L \approx 0.77 \mu \theta_L$. At thirty turns and $\mu = 0.3$ it peaks near 46, giving a ratio between the growing and decaying modes of $e^{46} \sim 10^{20}$, against the $\sim 10^{16}$ that double precision carries. The two-point boundary value problem in u is then genuinely unsolvable: at $x/L = 0.6$ with those parameters, successively refined grids give $F = 14.5, 41.1, -80.2, -26.3$ N — sign-flipping noise, not a converging sequence. This is a conditioning limit rather than a discretization error, and grid refinement does not touch it. The physically correct treatment in that regime is a one-sided boundary-layer problem near $u = L$ rather than a two-point problem across the whole ribbon; that has not been carried out. The parameters quoted in Section 9.2 keep $\lambda_+ L \approx 23$, which is solvable and converges.

A parameter squeeze. Two requirements pull against each other. Self-contact needs πr_{nat} comfortably below W (Section 5.1), which favours a small coil radius; conditioning needs $\mu L/r_{\text{nat}}$ small, which favours a large one. At $L = 200$ cm and $W = 8$ cm both can be met, but only with modest margin.

The energy cross-check is inadmissible. There is only one force, Eq. (99), and it is what the solver reports. The alternative of computing $\frac{dU_{\text{elastic}}}{dx}$ plus a Coulomb dissipation integral is not a second route to the same number but an inadmissible one, for the reason given in Section 8.5: it presumes the stored energy can be split into recoverable and friction-locked parts. Evaluated

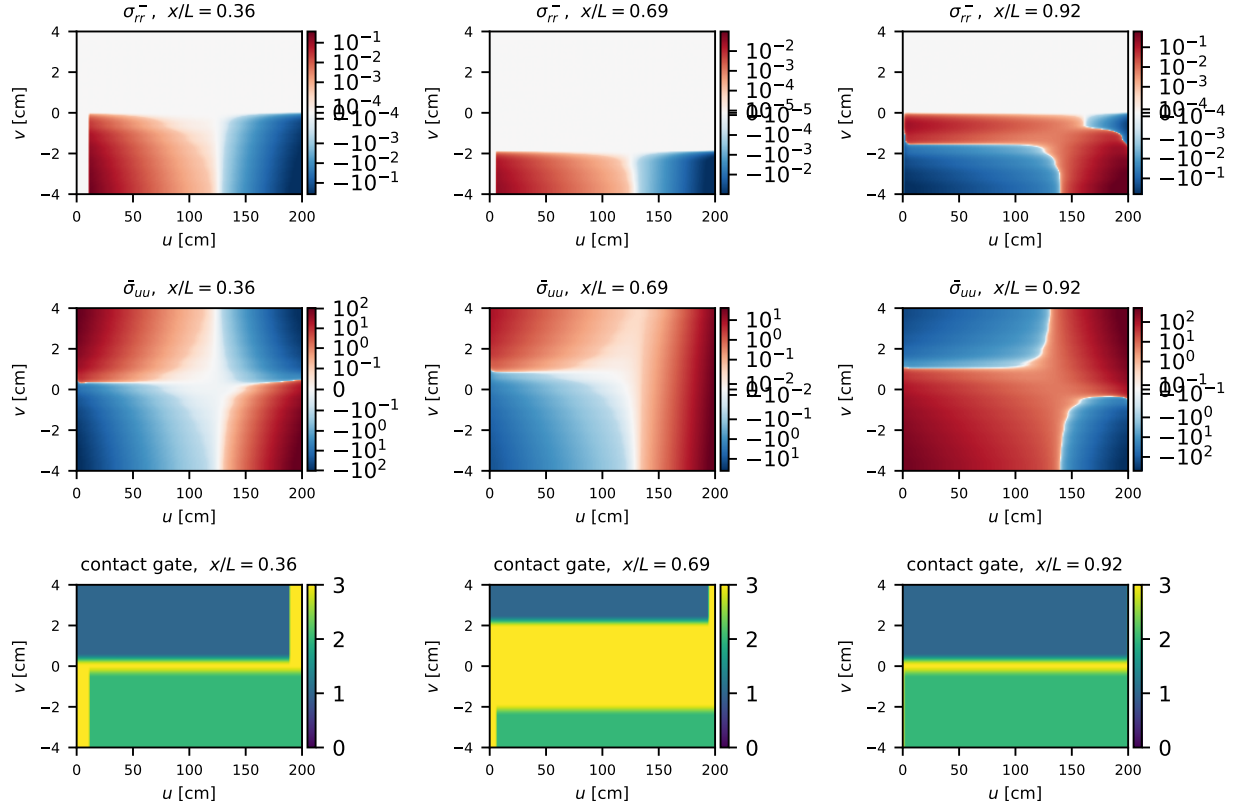


Figure 5: Stress fields at three extensions. Top: contact pressure σ_{rr}^- . Middle: in-plane $\bar{\sigma}_{uu}$, showing the boundary-layer structure implied by the amplification exponent Eq. (105) — stress concentrated within $1/\lambda_+$ of the ends, with a nearly stress-free interior. Bottom: the contact gates of Eqs. (40)–(41), sweeping across the width as the axial offset between turns grows.

regardless, it differs by factors of two to five. This is worth stating only so that the discrepancy is not mistaken for a numerical failure.

Unilateral contact is not enforced. Assumption 7 of Section 1.2 requires $\sigma_{rr} \leq 0$, but the governing equations contain no complementarity condition, and the solution does return tensile patches — up to about half the contact area at large extension. Doing this properly is a free-boundary problem in which contact is lost wherever the pressure would go positive, and it is not solved here. Since the friction magnitude depends on $|\sigma_{rr}|$, this is a genuine quantitative caveat on the friction branch, though it does not touch F_{bend} .

10 Predictions and experimental validation

The two branches of the model make predictions of very different quality, and the difference is worth stating plainly.

F_{bend} (Eq. 89) contains no adjustable parameter. It is monotonic, faster than exponential, and its magnitude is set by WD/r_{nat}^2 . For the geometry of Section 9.2 the soft-branch slope of

Eq. (90) evaluates to about 0.19 N, the same order as the prefactor of the slowly growing term in the measured fit of Figure 6. That is encouraging but not decisive, since a curve of roughly the right shape and magnitude can arise for more than one reason.

Two sharper tests follow from F_{bend} 's functional form, and both discriminate against a friction-dominated reading:

- **Independence of μ .** F_{bend} contains no friction coefficient at all. Changing the interface — a different paper stock, talc, graphite, a change in humidity — should leave $F(x)$ essentially unchanged if the bending branch dominates, and change it substantially otherwise.
- **Cubic thickness scaling.** $F_{\text{bend}} \propto D \propto t^3$, and also $\propto W$ and $\propto r_{\text{nat}}^{-2}$. Doubling the sheet thickness at fixed winding geometry should multiply the force by eight. A cubic law is hard to confuse with anything else, and this is probably the single most decisive measurement available.

The friction branch, by contrast, predicts a force proportional to μ and amplified by $e^{0.77\mu\theta_L}$, so it should be extremely sensitive to both the interface and the number of turns. The number of turns is easy to vary and is a clean second discriminator.

Finally, Section 7.2 predicts a sliding-to-stretching transition at $1 - x_*/L \sim (t/r_{\text{nat}})^{2/3}$, or at $x/L = \sqrt{1 - \rho_{\text{jam}}^2}$ if the coil jams first, with the stopping radius ρ_{stop} independent of the material. Measuring the coil radius at the point where sliding visibly stops, and comparing it with half the measured radial build over the initial radius, tests Eq. (??) directly.

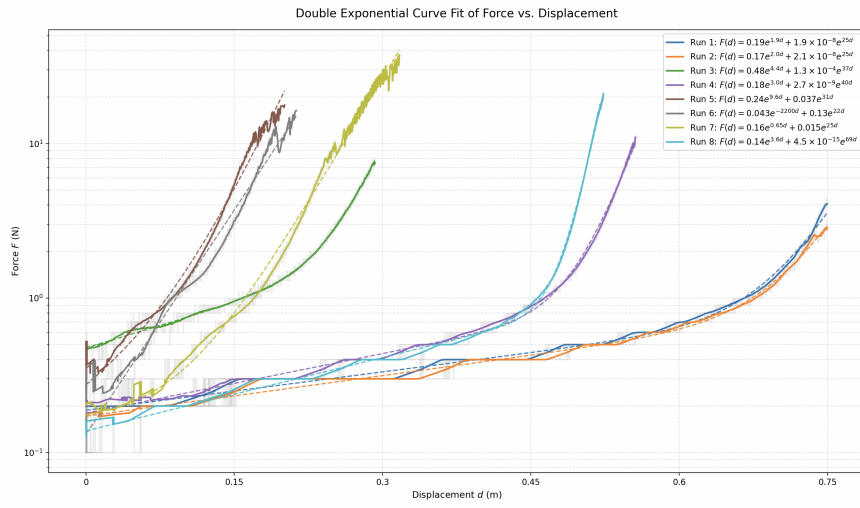


Figure 6: Instron force–displacement data: raw quantized data (grey), smoothed (solid), and curve fit (dashed).